Evaluation runs: **sikv** (kv-snapshot-isolation-tla)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	conforms	0.6		10 min, 16 GB	0.6	0.3	0.0	in-memory	brain05, 2026-11-02
3	48	90	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	384	2520	conforms	0.5	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
5	3840	113400	conforms	13.1	48	10 min, 16 GB	0.3	12.8	0.3	in-memory	brain05, 2026-11-02
6	46080	7484400	conforms	538.1	1138	10 min, 16 GB	4.0	529.2	8.8	in-memory	brain05, 2026-11-02

offline: pair **si**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	1	violation	0.6	8	10 min, 16 GB	0.6	0.0	0.0	in-memory	brain05, 2026-11-02
3	48	90	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	384	2520	41	violation	0.5	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
5	3840	113400	761	violation	11.8	34	10 min, 16 GB	0.3	11.6	0.2	in-memory	brain05, 2026-11-02
6	46080	7484400	20422	violation	493.6	685	10 min, 16 GB	4.0	487.3	6.4	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 7 si fifo, 7 serializable fifo.

skipped: a smaller size had reached its limit: 7 serializable fifo, 7 si fifo, 8 si fifo, 8 serializable fifo, 9 si fifo, 9 serializable fifo, 10 si fifo, 10 serializable fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	conforms	3.3	169	10 min, 16 GB	0.6	3.3	0.0	in-memory	brain01, 2026-11-02
3	48	90	conforms	3.5	160	10 min, 16 GB	0.1	3.5	0.0	in-memory	brain01, 2026-11-02
4	384	2520	conforms	9.2	794	10 min, 16 GB	0.1	9.2	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **si**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	1	violation	3.1	164	10 min, 16 GB	0.6	3.1	0.0	in-memory	brain01, 2026-11-02
3	48	90	4	violation	3.5	175	10 min, 16 GB	0.1	3.5	0.0	in-memory	brain01, 2026-11-02
4	384	2520	41	violation	9.2	746	10 min, 16 GB	0.1	9.2	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	conforms	3.1	160	10 min, 16 GB	3.1	0.3	0.0	in-memory	brain01, 2026-11-02
3	48	90	conforms	3.2	166	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-02
4	384	2520	conforms	3.9	164	10 min, 16 GB	3.8	0.4	0.1	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **si**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	6	1	violation	3.1	160	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	48	90	4	violation	3.2	166	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-02
4	384	2520	41	violation	3.8	164	10 min, 16 GB	3.8	0.4	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	6	conforms	3.1	156	1 h, 250 GB		brain01, 2026-11-02
3	48	90	conforms	3.2	162	1 h, 250 GB		brain01, 2026-11-02
4	384	2520	conforms	4.2	160	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **si**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	6	4	violation	3.1	156	1 h, 250 GB	1	brain01, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
3	48	90	48	violation	3.2	162	1 h, 250 GB	2	brain01, 2026-11-02
4	384	2520	1152	violation	4.2	160	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.4	169	10 min, 16 GB	3.3	0.3	0.0	in-memory	brain01, 2026-11-02
3	90	90	conforms	3.6	160	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-02
4	2520	2520	conforms	9.2	794	10 min, 16 GB	9.2	0.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.1	164	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	90	90	conforms	3.5	175	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-02
4	2520	2520	conforms	9.2	746	10 min, 16 GB	9.2	0.4	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.3	169	10 min, 16 GB	0.3	3.3	0.0	in-memory	brain01, 2026-11-02
3	90	90	conforms	3.5	160	10 min, 16 GB	0.1	3.5	0.0	in-memory	brain01, 2026-11-02
4	2520	2520	conforms	9.2	794	10 min, 16 GB	0.4	9.2	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.1	164	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	90	90	conforms	3.5	175	10 min, 16 GB	0.1	3.5	0.0	in-memory	brain01, 2026-11-02
4	2520	2520	conforms	9.2	746	10 min, 16 GB	0.4	9.2	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.1	162	10 min, 16 GB	3.1	0.7	0.0	in-memory	brain02, 2026-11-03
3	90	90	conforms	3.8	168	10 min, 16 GB	3.7	0.2	0.1	in-memory	brain02, 2026-11-03
4	2520	2520	conforms	9.1	778	10 min, 16 GB	9.1	0.5	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.6	172	10 min, 16 GB	3.6	0.3	0.0	in-memory	brain02, 2026-11-03
3	90	90	conforms	3.9	163	10 min, 16 GB	3.9	0.1	0.0	in-memory	brain02, 2026-11-03
4	2520	2520	conforms	10.1	862	10 min, 16 GB	9.9	0.4	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.1	162	10 min, 16 GB	0.7	3.1	0.0	in-memory	brain02, 2026-11-03
3	90	90	conforms	3.8	168	10 min, 16 GB	0.2	3.7	0.1	in-memory	brain02, 2026-11-03
4	2520	2520	conforms	9.1	778	10 min, 16 GB	0.5	9.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	6	conforms	3.6	172	10 min, 16 GB	0.3	3.6	0.0	in-memory	brain02, 2026-11-03
3	90	90	conforms	3.9	163	10 min, 16 GB	0.1	3.9	0.0	in-memory	brain02, 2026-11-03
4	2520	2520	conforms	10.0	862	10 min, 16 GB	0.4	9.9	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 fifo serializable, 5 fifo si.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	6	conforms	4.0	167	1 h, 250 GB		brain02, 2026-11-03
3	90	90	conforms	4.2	177	1 h, 250 GB		brain02, 2026-11-03
4	2520	2520	conforms	11.0	4789	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	6	conforms	4.9	162	1 h, 250 GB		brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3	90	90	conforms	4.6	162	1 h, 250 GB		brain02, 2026-11-03
4	2520	2520	conforms	12.1	4779	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **serializable**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	6	conforms	3.3	158	1 h, 250 GB		brain02, 2026-11-03
3	90	90	conforms	3.8	164	1 h, 250 GB		brain02, 2026-11-03
4	2520	2520	conforms	9.6	774	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **si**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	6	conforms	3.6	167	1 h, 250 GB		brain02, 2026-11-03
3	90	90	conforms	3.9	159	1 h, 250 GB		brain02, 2026-11-03
4	2520	2520	conforms	10.5	857	1 h, 250 GB		brain02, 2026-11-03